



KENYATTA UNIVERSITY

EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE

(GAS AND PETROLEUM ENGINEERING)

EPL 102: COMPUTER PROGRAMMING I

1ST SEMESTER 2021/2022

TIME: 2 HOURS

INSTRUCTIONS:

- This paper consists of FIVE Questions.
- Question ONE is compulsory and contains 30 marks
- Attempt any TWO of the remaining FOUR Questions. Each Question is 20 marks.

60-70 E
50-60 C
40-50 = P

QUESTION 1: (30 MARKS)

- (a) Define the following terms and give example for each

- High level languages
- Semantic error

[4 marks]

- (b) Design a program to compute dew point value as represented in 1 C below
[6 marks]

The dew point temperature T_d can be calculated (approximately) from the relative humidity (RH) and the actual temperature T by

$$T_d = \frac{b \cdot f(T, RH)}{a - f(T, RH)}$$

$$f(T, RH) = \frac{a \cdot T}{b + T} + \ln(RH)$$

$$T_d = \frac{b \cdot \left(\frac{a \cdot T}{b + T} + \ln(RH) \right)}{a - \left(\frac{a \cdot T}{b + T} + \ln(RH) \right)}$$

where $a = 17.27$ and $b = 237.7^\circ \text{C}$.

Write a program that reads the relative humidity (between 0 and 1) and the temperature (in degrees C) and prints the dew point value. Use the C++ function `log` to compute the natural logarithm.

[8 marks]

if include $\sqrt{}$
if include $\cos$
using an old
int main

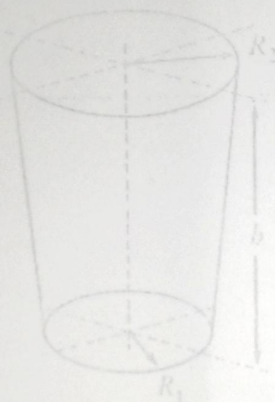
- (d) Write a C++ program that computes and display volume of oil cylindrical containers of height 10 metres and radius between 2.5 metres and 6.7 metres . Program should display volume in cubic meter.

$$V = \pi r^2 h$$

[6 marks]

(e)

A laboratory container is shaped like the frustum of a cone:



Write functions to compute the volume and surface area, using these equations:

$$V = \frac{1}{3} \pi b (R_1^2 + R_2^2 + R_1 R_2)$$

$$S = \pi (R_1 + R_2) \sqrt{(R_2 - R_1)^2 + b^2} + \pi R_1^2$$

6

[8 marks]

Question 2(20 Marks).

- (a) Using a simple diagram illustrate structure of do--- while loop. [3 marks]
(b) Briefly explain stages of computer problem solving. State product of each stage.

[4 marks]

- (c) Design a program to compute a car drag force using formula in 2 d below

[6 marks]

(d)

The drag force on a car is given by

$$F_D = \frac{1}{2} \rho v^2 A C_D$$

where ρ is the density of air (1.23 kg/m^3), v is the velocity in units of m/s, A is the projected area of the car (2.5 m^2), and C_D is the drag coefficient (0.2).

The amount of power in watts required to overcome such drag force is $P = F_D v$, and the equivalent horsepower required is $\text{Hp} = P / 746$. Write a program that accepts a car's velocity and computes the power in watts and in horsepower needed to overcome the resulting drag force. Note: $1 \text{ mph} = 0.447 \text{ m/s}$.

[7 marks]

Write a program to display various MC and C++ statements between 0 to 100 in steps of 3 (8 marks)

Question 5(28 Marks)

Outline main stages in problem solving using a computer (4 marks)

Define the following terms

- 1. Array
- 2. Structure
- 3. Pointer

4. function implementation

polys
casting
library and library
operation

(8 marks)

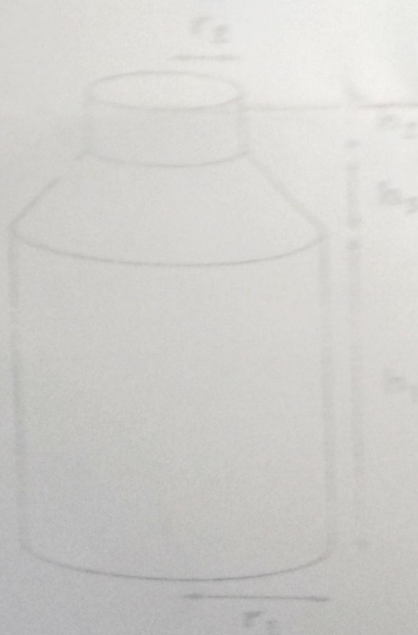
Diagram below represent shape of a bottle

The shape of a bottle is approximated to two cylinders of radius r_1 and r_2 and heights h_1 and h_2 joined by a cone section of height h_3 .

Using the known fact for the volume of a cylinder $V = \pi r^2 h$ and cone section,

$$V = \pi \frac{(r_1^2 + r_1 r_2 + r_2^2) h_3}{3}$$

Write a program to compute volume V (3 marks)



#include <iostream>
#include <math>
using namespace std;
int main()

```
{
    double r1, r2, h1, h2, h3, V;
    cout << "Enter radius of bottom cylinder: ";
    cin >> r1;
    cout << "Enter height of bottom cylinder: ";
    cin >> h1;
    V1 = M_PI * r1 * r1 * h1;
    cout << "Volume of bottom cylinder is: ";
    cout << V1 << " cm^3\n";
}
```

END OF EXAMINATION